Data Cleaning & Preprocessing Summary

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1. Cleaning Scope & Overview

Following the initial audit conducted by Nourhan, this stage focuses on cleaning and preparing the car_prices.csv dataset in Microsoft Excel. The goal is to eliminate duplicate entries based on unique vehicle identifiers, handle missing values, and produce a clean, reliable dataset ready for SQL modeling and Power BI calculations.

Initial Rows	Duplicates Removed	Final Clean Records	Final File Format
302,021	3,719	298,302	Excel Workbook (.xlsx)

2. Cleaning Steps & Methodology

The dataset was processed and sanitized using Microsoft Excel following these clear steps:

Cleaning Action	Method Used in Excel	Result & Impact
Deduplication by VIN	Data Tab -> Remove Duplicates on VIN column	Removed 3,719 duplicate vehicle records to ensure each car is counted uniquely.
Blank & Null Handling	Find & Select -> Go To Special -> Blanks	Filtered out incomplete records with missing critical values to prevent skewed calculations.
Data Type Standardization	Format Cells (General, Text, Number, Date)	Ensured year and prices are formatted properly as numbers and categorical text is clean.
File Migration to XLSX	Save As -> Excel Workbook (.xlsx)	Safeguarded data structure and formatting from text truncation associated with raw CSV.

3. Cleaning Results & Summary

Cleaning Metric	Record Count / Status	Note for Next Phase
Duplicate Records Removed	3,719 rows	Eliminated repeated VIN occurrences to avoid double counting sales.
Clean & Unique Records Retained	298,302 rows	Solid baseline of verified car sale transactions.
Data Quality Status	100% Cleaned & Verified	Ready for SQL relational schema creation and KPI measures.

4. Handover to Data Modeling & Calculation Team

The cleaned workbook is saved as car_prices_cleaned.csv final.xlsx (298,302 clean rows) and handed over to:

- 1. Mohammed Alaa (Data Modeling): To import into the SQL database, create primary and foreign keys, and construct the Star Schema.
- 2. Hossam Hisham (Data Calculation): To write SQL analytical queries, measure aggregations, and prepare DAX calculations for Power BI.